SMART AGRICULTURE USING THE INTERNET OF THINGS (IoT)

CodeAlpha Internship — Task 1: Research Report

Submitted by: Anusha G
Branch: Electronics and Communication Engineering (ECE)
Internship Domain: Internet of Things (IoT)

October 2026

Abstract

Smart agriculture uses digital technologies to monitor crops, soil, weather and farm equipment. The Internet of Things (IoT) connects sensors and devices so that information can be collected and shared through a network. Farmers can use this information to make better decisions about irrigation, crop health and resource use. This report explains the basic working of IoT in agriculture, its main components, applications, benefits, challenges and future scope. A smart irrigation example shows how sensor readings can help farmers decide when crops need water.

1. Introduction

Agriculture depends on soil quality, water availability, weather and timely farm operations. Traditional farming knowledge remains valuable, but checking every part of a large field manually can take time and may not reveal changing conditions quickly. Smart agriculture, also called smart farming, combines agricultural practices with technologies such as sensors, wireless communication, cloud computing and data analysis. The Internet of Things (IoT) allows physical devices to collect data and exchange it over a network. In farming, these connected devices can help farmers observe field conditions and respond more accurately. IoT supports decision-making; it does not replace farmers' experience or guarantee higher yields in every situation.

2. Main Components and Working Principle

An IoT-based agricultural system usually contains five main parts. First, sensors measure conditions such as soil moisture, temperature, humidity, light intensity, water level or soil pH. Second, a controller such as an Arduino, ESP32 or Raspberry Pi reads the sensor values. Third, a communication method—such as Wi-Fi, cellular networks, Bluetooth or a low-power long-range network—sends data to a phone, computer or online platform. Fourth, software or a dashboard displays readings and may compare them with selected limits. Finally, an actuator such as a relay-controlled pump or valve can perform an action when the system is configured for automatic control. The process begins when a sensor measures a field condition. The controller processes the reading and sends it to a monitoring platform. A farmer can view the data and decide what action is needed. In an automated system, the controller may switch on irrigation when soil moisture falls below a chosen threshold and switch it off when sufficient moisture is reached. The threshold must be selected for the crop, soil and local conditions; one setting is not suitable for every farm. Reliable systems also need sensor calibration, suitable power, weather-resistant installation and a way to handle network interruptions.

3. Applications of IoT in Smart Agriculture

Smart irrigation is one important application. Soil-moisture sensors help estimate whether the root zone is becoming dry. Combined with weather information and irrigation scheduling, these readings can support more careful watering and reduce unnecessary pumping. Flow or water-level sensors can also help identify abnormal water use or an empty storage tank. Crop and soil monitoring is another application. Temperature, humidity, light and soil sensors provide regular information about the field environment. Farmers can use the readings to identify unusual conditions and decide when to inspect plants. Soil pH and nutrient-related measurements may support soil management, although specialist testing can still be necessary. IoT can also support greenhouse management. Sensors can monitor temperature and humidity, while controllers operate fans, ventilation or irrigation equipment according to configured rules. In livestock farming, connected devices may monitor water availability, location or selected animal-health indicators. In storage facilities, temperature and humidity monitoring can help operators identify conditions that may damage

harvested produce. These applications show how timely data can assist farm management from planting to storage.

4. Advantages

IoT can make field information available more frequently than occasional manual checks. Better information can help farmers plan irrigation, reduce unnecessary field visits and use water, electricity and other inputs more carefully. Alerts can draw attention to unusual readings, allowing a farmer to investigate earlier. Recorded data can also help compare conditions over time and improve future planning. Remote monitoring may be useful when fields are far from the farmer's home or when several plots need attention. The actual benefits depend on system design, crop type, local conditions, maintenance and how well the information is used.

5. Challenges and Limitations

The initial cost of sensors, controllers, communication equipment and installation may be difficult for small farms to afford. Some rural areas have weak internet or mobile coverage, and devices in fields may have limited access to reliable electricity. Sensors can produce inaccurate readings if they are poorly installed, not calibrated or damaged by water, heat, dust or farm machinery. Systems also require maintenance and basic technical knowledge. Data security and privacy are important because connected devices and online accounts may be exposed to misuse if passwords, software and network settings are not managed properly. Automated equipment should include safe manual controls so that a faulty sensor or connection does not leave a pump running unnecessarily. Farmers should evaluate the cost, repair options, local support and expected usefulness before adopting a system. Simple, affordable designs and training can make the technology more accessible.

6. Future Scope

Future smart farms may combine IoT with artificial intelligence, satellite images, drones and weather forecasts. AI tools may help identify patterns in sensor data, estimate irrigation needs or flag possible crop stress, but their predictions depend on data quality and local validation. Solar-powered devices and low-power communication may improve operation in remote fields. More affordable sensors, easy-to-understand local-language applications and farmer training could support wider adoption. Interoperable systems that allow devices from different manufacturers to work together would also make it easier to expand or repair a farm installation.

7. Conclusion

The Internet of Things can support smart agriculture by connecting sensors, controllers, communication networks and monitoring software. It helps farmers observe soil and weather conditions, manage irrigation, monitor greenhouses and respond to selected farm problems. Its main value is timely information that supports practical decisions. However, cost, connectivity, power, sensor accuracy, maintenance and security must be considered. IoT works best when technology is selected for the farm's actual needs and used alongside agricultural knowledge. With suitable design and training, smart agriculture can contribute to more informed and efficient farm management.

References

[1] García, L., Parra, L., Jimenez, J. M., Lloret, J., and Lorenz, P. (2020). "IoT-Based Smart Irrigation Systems: An Overview on the Recent Trends on Sensors and IoT Systems for Irrigation in Precision Agriculture." Sensors, 20(4), 1042. https://doi.org/10.3390/s20041042

[2] Kumar, S. Vinod, Singh, C. D., and Upendar, K. (2020). “Review on IoT Based Precision Irrigation System in Agriculture.” Current Journal of Applied Science and Technology, 39(45), 15–26. https://doi.org/10.9734/cjast/2020/v39i4531156

[3] He, Y. et al. (2024). “Smart Sensors and Smart Data for Precision Agriculture: A Review.” Sensors, 24(8), 2647. https://doi.org/10.3390/s24082647

[4] Moustafa, A., Abdelwahab, O. M. M., Ricci, G. F., and Gentile, F. (2026). “Internet of Things-enabled smart irrigation systems for precision water management: A systematic review.” Agricultural Water Management, 333, 110615. https://doi.org/10.1016/j.agwat.2026.110615